CS380 Midterm Study Guide

Introduction to AI

- **Potential Goals of AI**:
  - Acting Humanly
  - Thinking Humanly
  - Acting Rationally
  - Thinking Rationally
- **Turing Test**:
  - Evaluates a machine's ability to exhibit intelligent behavior indistinguishable from a human.
- **Rational Agents**:
  - **Agent**: An entity that perceives and acts.
  - **Percepts**: The agent’s perceptual inputs.
  - **Actions**: What the agent can do.
  - **Sensors**: Devices to perceive the environment.
  - **Actuators**: Devices to act in the environment.
  - **Rational Agent**: Selects actions to maximize expected performance based on percept sequence and built-in knowledge.
- **PEAS Framework**:
  - Stands for: Performance measure, Environment, Actuators, Sensors.
  - Helps define rational agents clearly.
  - Example: **Automated Taxi**
    - Performance: Safe, fast, legal, comfortable trip.
    - Environment: Roads, traffic, passengers.
    - Actuators: Steering, accelerator, brake.
    - Sensors: Cameras, GPS, speedometer.
- **Environment Types**:
  - **Observable** vs. **Partially Observable**
  - **Deterministic** vs. **Stochastic**
  - **Episodic** vs. **Sequential**
  - **Static** vs. **Dynamic**
  - **Discrete** vs. **Continuous**
  - **Single-Agent** vs. **Multi-Agent**

Problem Solving & Uninformed Search

- **Problem Solving Concepts**:
  - A problem is defined by:
    - Initial state
    - Actions
    - Transition model
    - Goal test

        - Path cost

- **State Space**:
    - The set of all states reachable from the initial state by any sequence of actions.
- **Random Walk**:
    - A method to randomly traverse a state space; often used as a baseline.
- **Uninformed Search Algorithms**:
    - **Breadth-First Search (BFS)**:
        - Explores all nodes at current depth before moving to the next.
        - Complete, Optimal (if cost = 1), Time: O(b^d), Space: O(b^d)
    - **Depth-First Search (DFS)**:
        - Explores as far down a branch as possible before backtracking.
        - Not complete (in infinite spaces), Not optimal, Time: O(b^m), Space: O(bm)
    - **Iterative Deepening**:
        - Repeated DFS with increasing depth limit.
        - Combines space efficiency of DFS and optimality of BFS.

Informed Search, A*, & Heuristics

- **Informed Search**:
    - Uses problem-specific knowledge to find solutions more efficiently.
- **Heuristics**:
    - Estimate of the cost to reach the goal from a node.
- **Greedy Best-First Search**:
    - Expands node that appears closest to goal (uses only heuristic h(n)).
- **A* Search**:
    - Uses: f(n) = g(n) + h(n)
        - g(n): Cost so far to reach node
        - h(n): Estimated cost to goal
    - Complete and optimal if h(n) is **admissible**.
- **Admissible Heuristic**:
    - Never overestimates the cost to reach the goal.

Local Search

- **Local Search & Iterative Improvement**:
    - Focuses on a single current state and moves to a neighbor state.
- **Hill Climbing**:

    - Selects the neighbor with the highest value (greedy).
    - Gets stuck in local maxima, ridges, plateaus.
- **Simulated Annealing**:
    - Accepts worse solutions with a probability that decreases over time.
    - Helps escape local maxima.

Adversarial Search

- **Minimax Algorithm**:
    - Used for deterministic, turn-based, zero-sum games.
    - Computes the optimal move by assuming opponent plays optimally.
- **Alpha-Beta Pruning**:
    - Prunes branches that cannot affect the final decision.
    - Maintains optimality of minimax but is more efficient.
- **Expecti-Minimax**:
    - Used for games with chance (e.g., dice).
    - Replaces min nodes with expectation nodes.
- **Monte Carlo Methods**:
    - Simulate many random games from a position to estimate value.

Logical Agents

- **Declarative Programming**:
    - Programming by specifying *what* should be achieved rather than *how*.
- **Wumpus World**:
    - Demonstrates inference and reasoning with uncertainty.
    - Agent uses percepts to build a knowledge base.
- **Propositional Logic**:
    - Syntax includes: symbols, logical connectives (¬, □, □, □, □)
    - Limited to facts, no variables.
- **First-Order Logic (FOL)**:
    - Includes:
        - **Objects**: Things in the domain.
        - **Relations**: Properties or connections (e.g., `Loves(John, Mary)`).
        - **Functions**: Map objects to other objects.
    - Quantifiers:
        - **Universal (□)**: For all.
        - **Existential (□)**: There exists.
    - Translate English to FOL:
        - "All humans are mortal" → □x (Human(x) □ Mortal(x))
        - "Some humans are philosophers" → □x (Human(x) □ Philosopher(x))